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Enhancing reliability in power grid fault management via multi-task learning-based knowledge graph completion

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Abstract

Timely detection and resolution of power equipment failures are vital for ensuring the stability and reliability of electrical grids. This paper presents a knowledge graph completion model (KGCM) that integrates multi-task learning with pre-trained models to enhance fault-handling knowledge graph representations. Entities and relations are encoded with a pre-trained model and refined through Translating Embeddings (TransE)-based structural constraints, while a Siamese network is introduced to improve feature extraction and robustness by predicting missing triplet components. The proposed framework was validated on a dataset from the State Grid Tianjin as well as standard benchmarks. Experimental results show that KGCM consistently outperforms the baseline Language Model Knowledge Embedding with TransE (LMKE-TransE) model in terms of Mean Reciprocal Rank (MRR) and Hits at N (Hits@N) metrics. In practical fault-action prediction, KGCM demonstrates superior accuracy, enabling more reliable diagnosis and handling of power grid faults. These findings confirm the effectiveness of combining pre-trained models, structural fine-tuning, and multi-task feature extraction for knowledge graph completion, and highlight the model's potential to enhance automation and dependability in power grid fault management.

Keywords Knowledge graph completion, Multi-task, Pre-trained models, TransE, Siamese network, Reliability, Robustness, Dependable systems

1 Introduction

The power system serves as a fundamental pillar of modern society, with its continuous operation being critical for national economic stability and societal well-being. The expansion of the grid and the enhancement of operational flexibility have increased the complexity of fault detection, posing significant challenges to the reliability and safety of power infrastructure. Urban power grids, which are essential for economic activities, can incur substantial economic losses and disruptions in the event of faults. Traditional fault detection methods, which rely on manual intervention, often struggle to accurately



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assess equipment states amidst complex failure scenarios [1, 2]. Therefore, more intelligent and automated fault detection approaches must be developed to manage the increasing complexity and variability of power systems [3, 4]. Such advancements can improve reliability, safety, operational efficiency, and contribute to sustainable development. Specifically, the proposed method enhances system reliability and robustness in practical applications by leveraging multi-task learning and TransE-based structural fine-tuning to ensure accurate fault knowledge representation, while the Siamese network improves feature discrimination for fault detection.

Contemporary power fault detection algorithms are generally categorized into two types: those based on conventional methodologies and those employing deep learning techniques [5, 6]. Conventional approaches, dependent on manual detection and intervention, may not effectively discern equipment states or provide timely responses during abrupt faults [7, 8]. The integration of knowledge graphs and artificial intelligence offers a promising avenue for leveraging deep learning to enhance fault detection and prediction while utilizing knowledge graphs for guidance [9, 10]. However, the effectiveness of existing models that combine reinforcement learning with knowledge graph-based fault handling is contingent upon the completeness and accuracy of the knowledge graph. Inaccuracies or incompleteness in the graph, such as erroneous or missing edges, necessitate refinement. Therefore, the development of an intelligently curated knowledge graph is crucial for proficient fault handling [11, 12].

Knowledge graphs are widely used in deep learning frameworks, particularly those that emphasize interpretability and causal path elucidation. Nonetheless, current knowledge graph completion algorithms primarily focus on improving representations without fully leveraging latent edges or correcting redundant ones, which may impact the accuracy of fault detection and handling. Additionally, while numerous studies incorporate pre-trained large language models for knowledge graph representation [13], these models often lack domain-specific structured knowledge constraints and may exhibit robustness issues in feature extraction. Hence, a framework that strengthens domain-specific knowledge graph representation and enhances feature extraction robustness is essential for improving the accuracy of fault handling automation.

This paper proposes a Multi-Task Knowledge Graph Completion Framework (KGCM) designed to improve the effective representation of fault-handling knowledge graphs and enhance the robustness of feature extraction networks. KGCM refines and augments pre-trained model knowledge graph representations based on domain-specific knowledge constraints using TransE and employs Siamese neural networks to enhance feature extraction robustness in predicting unknown triplet information. By doing so, KGCM enriches the representation of fault-handling knowledge graphs, thereby improving the reliability of downstream automated fault detection and handling.

The contributions of this study are as follows:

Enhancement of fault-handling knowledge graph representation through the integration of domain-specific knowledge constraints and pre-trained models.

Improvement of feature extraction capabilities in predicting triplet processes within fault-handling knowledge graphs using Siamese neural networks.

Application of the multi-task learning-based knowledge graph completion framework to fault-handling knowledge graph representation and completion tasks, providing a robust foundation for downstream applications.

The use of multi-task learning enhances the accuracy and robustness of embedding models, aiding in the correction of node and edge representations within knowledge graphs. This framework not only offers a superior foundation for subsequent tasks but also has valuable applications in various power grid operation and maintenance scenarios, providing insights into fault diagnosis and handling. Distinct from existing approaches that merely stack language models with knowledge embeddings (such as standard BERT + TransE or LMKE), the proposed KGCM framework is specifically engineered to address the high-risk and long-tail distribution characteristics of power grid fault data. The core innovation lies in the introduction of a Siamese network as a "Fine-grained Verifier" within the feature space. While pre-trained models offer rich semantic priors, they often lack discriminative power when facing sparse and highly similar fault samples in the grid. By leveraging a multi-task learning mechanism, KGCM enforces a rigorous similarity verification via the Siamese network alongside semantic completion. This design effectively distinguishes confusing fault patterns, thereby solving the issue of insufficient robustness in traditional models on long-tail data and ensuring the high reliability required for safety-critical power systems.

The remainder of this paper is organized as follows: Sect. 2 reviews related work; Sect. 3 details the design of the multi-task knowledge graph completion model; Sect. 4 presents experimental settings and analyses; and Sect. 5 concludes the paper and outlines future research directions.

2 Related works

2.1 Knowledge graph completion

Knowledge graph completion is a critical task in knowledge representation and reasoning, garnering significant attention in recent years. Various methodologies and techniques have been proposed to address this task. Meilicke et al. [14] introduced an anytime bottom-up technique that excels in learning logical rules from extensive knowledge graphs, outperforming conventional rule-based approaches. Another notable advancement is Knowledge Graph BERT(KG-BERT), proposed by Yao et al. [15], which utilizes pre-trained language models to represent triples in knowledge graphs as textual sequences. Similarly, Zhang et al. [16] introduced hypercomplex representations for knowledge graph embeddings, achieving state-of-the-art performance on established benchmarks. Knowledge acquisition, particularly in the context of knowledge graph completion, has been a focal point of research. Ji et al. [17] conducted a comprehensive review of techniques, including embedding methods, path inference, and logical rule reasoning. Recently, Zhang et al. [18] proposed a Relational Graph Neural Network with Hierarchical Attention, aimed at inferring missing values in incomplete knowledge triples. Addressing the limitations of existing graph embedding approaches, Wang et al. [19] enhanced textual encoding with graph embedding techniques to achieve more efficient knowledge graph completion. Additionally, Zhang et al. [20] presented a neural network-based approach for drug repurposing for COVID-19, leveraging knowledge graph completion techniques. Overall, knowledge graph completion is crucial in knowledge representation and reasoning, driving the development of innovative approaches to improve performance and accuracy in inferring missing values.

2.2 Power grid fault diagnosis based on knowledge graph

The integration of knowledge graph technology into power grid equipment fault diagnosis has gained considerable attention in recent years. Yun et al. [21] focused on developing a knowledge question-and-answer system for diagnosing faults in the Electric Information Collection System, leveraging knowledge graph technology to enhance reasoning efficiency and support the intellectualization of fault diagnosis. Huang et al. [22] emphasized the construction of knowledge graphs for large-scale power grids in China to improve operational, maintenance, and management efficiency. Bo et al. [23] demonstrated the effectiveness of a method based on understanding fault disposal preplan knowledge, reducing fault disposal pressure and improving efficiency. Knowledge graph technology is also applied to fault diagnosis within specific components of the power grid system. For example, Chan et al. [24] explored power oscillation detection and classification using telemetry data monitoring systems based on graph theory. Li et al. [25] introduced a dynamic graph-based anomaly detection algorithm for the electrical grid, enabling the rapid identification of unusual events through sensor data analysis. Zhou et al. [26] developed a fault model for power communication equipment and constructed an equipment fault knowledge graph for visual representation. Moreover, Wu et al. [27] proposed a knowledge graph technology-based fault diagnosis framework for HVDC transmission systems to address limitations in existing fault diagnosis methods. Liu et al. [28] detailed the application of knowledge graphs in smart grid fault diagnosis, emphasizing knowledge extraction, fusion, processing, and the construction of fault scheduling knowledge graphs for effective fault diagnosis systems. In summary, the literature review highlights the growing interest and application of knowledge graph technology in enhancing fault diagnosis processes for power grid equipment.

Furthermore, recent research increasingly focuses on integrating explicit domain knowledge and domain adaptation techniques to address data scarcity and distribution shifts in fault diagnosis. For instance, Ren et al. proposed a Physics-guided TL-LSTM network for lithium-ion battery degradation prediction, effectively combining physical laws with transfer learning to enhance generalization capabilities [41]. Similarly, a comprehensive survey on domain adaptation for intelligent fault diagnosis emphasized the critical role of transferring diagnostic knowledge across varying working conditions and equipment types [42, 43]. Drawing inspiration from these domain-adaptive paradigms, our KGCM framework employs the knowledge graph itself as a structured domain constraint. Unlike traditional statistical adaptation, we utilize multi-task learning to align the semantic representations of long-tail fault samples with the verified topological structure of the grid, effectively solving the domain adaptation challenge in reliable fault handling.

2.3 Siamese network

Siamese networks have been extensively applied across various domains, including visual object tracking, text semantic understanding, change detection in remote sensing imagery, and entity alignment. In visual object tracking, Li et al. [29] introduced a gradient-guided network designed to enhance Siamese network performance by leveraging discriminative gradient information. In the field of natural language processing, Tang et al. [30] demonstrated the effectiveness of Bidirectional Long Short-Term Memory Networks (BiLSTM) in improving text semantic understanding. In the context

of change detection in remote sensing imagery, Touati et al. [31] and Jiang et al. [32] contributed by developing partially uncoupled Siamese models and pyramid feature-based attention-guided Siamese networks, respectively. Zou et al. [33] and Yan et al. [34] further advanced the field with deeper Siamese networks incorporating spatiotemporal enhancement and two-level feature fusion for real-time object tracking and remote sensing image change detection. For entity alignment, Chen et al. [35] introduced a multi-modal Siamese network that effectively leverages multi-modal knowledge. To address the challenges associated with complex network architectures and high computational costs in change detection tasks, Huang et al. [36] proposed spatiotemporal enhancement and interlevel fusion networks. Overall, these studies highlight the significance of feature enhancement and capture in Siamese networks across diverse applications, contributing to improved performance and accuracy.

3 Proposed model

3.1 Framework

This section presents a framework for knowledge graph completion in power grid fault handling based on multi-task learning, as illustrated in Fig. 1.

The proposed KGCM framework is not a simple concatenation of pre-trained models and structured embedding modules; rather, it is a multi-task synergistic system designed to tackle the challenge of long-tail distributions in power equipment faults. As illustrated in Fig. 1, this framework advances beyond existing methods like LMKE [37] by fundamentally optimizing their discriminatory capability for low-frequency, high-risk faults. Our core design philosophy operates on two levels: First, we employ the joint encoding of pre-trained language models (BERT) and TransE to mitigate knowledge graph sparsity, providing global semantic and structural constraints. Second, and most critically, we introduce a parallel Siamese network module to perform fine-grained verification in the feature space. Traditional LMKE models rely on MLP-based contrastive learning, which performs well with ample samples but tends to misclassify long-tail fault samples due to blurred feature boundaries. KGCM injects the strong discriminatory power of the Siamese network into the embedding process via multi-task learning, compelling the model to perform a rigorous "semantic-structural-feature" triple verification during

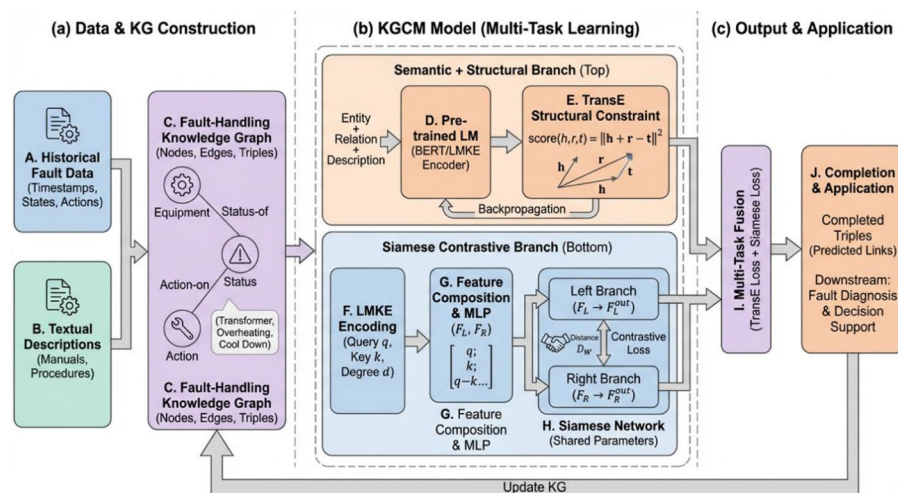


Fig. 1 Knowledge graph completion framework for grid fault handling based on multi-task learning

triplet prediction. This mechanism significantly enhances robustness in complex fault scenarios, ensuring the accuracy of automated fault handling recommendations.

The following sections will detail three key components of the framework. First, the foundational LMKE model utilized in this study will be introduced. Second, the use of the TransE structured encoding network combined with pre-trained models for knowledge graph representation will be explained. Finally, the integration of Siamese neural networks to improve the feature extraction module will be discussed. In this framework, the large language model processes the initial representations of entities and relations within the triplets, incorporating both pre-trained knowledge and descriptive information. This approach generates encoded vectors h , r , and t in R^d , integrating entity and relation descriptions with other context-related information from BERT, thus enhancing the overall representation beyond intrinsic data.

LMKE outperforms previous models by not only improving the effective representation of triplets through the use of large language models and additional triplet descriptions but also by refining negative sampling strategies. Unlike earlier approaches, LMKE treats the triplet itself as the positive sample within the current batch, while the known data of the triplet and other targets from the batch, excluding the positive target, are used as negative samples. This novel methodology optimizes the negative sampling and encoding processes, enhancing the model framework's efficiency. For link prediction, LMKE replaces traditional cosine similarity-based calculations with a multi-layer perceptron for measuring compatibility. Additionally, to enhance the distinction between entity representations, the model includes features such as node degrees during the encoding of known data features. These refinements improve the scientific rigor and accuracy of triplet representations within the knowledge graph.

To further highlight the technical innovations, we summarize the key differences between the baseline LMKE and our proposed KGCM framework. Table 1 clearly outlines the improvements in terms of input representation, structural constraints, feature extraction, and negative sampling strategy.

As shown in Table 1, KGCM not only builds upon the strengths of LMKE but also introduces multi-task structural constraints and Siamese feature extraction, thereby achieving more robust and domain-adaptive knowledge graph completion.

3.2 Fine-tuning based on TransE

This module focuses on refining graph representation through fine-tuning based on the TransE model, as illustrated in Fig. 2.

Table 1 Comparison between LMKE and the proposed KGCM framework

Aspect	LMKE	KGCM (Proposed)
Input representation	Uses pre-trained language models with basic triplet textual descriptions	Enriches triplets with pre-trained models plus auxiliary textual descriptions and entity degree features, improving long-tail entity representation
Structural constraints	Limited structural refinement; relies mainly on semantic encoding	Integrates TransE-based fine-tuning to enforce domain-specific structural and relational constraints in knowledge graphs
Feature extraction	Employs MLP-based contrastive learning for similarity measurement	Introduces a Siamese network for robust feature extraction, enhancing discrimination between similar triplets

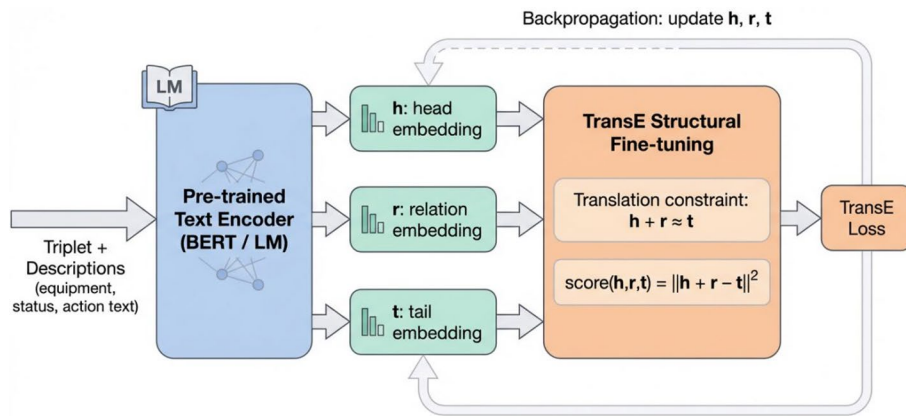


Fig. 2 Architecture for fine-tuning the pre-trained model based on TransE

After the pre-trained model generates entity features and relationships for the triplets, TransE is utilized to fine-tune these representations, enhancing their domain-specific and structural semantics.

The objective of this module is to enhance the encoding of triplets derived from pre-trained large-scale models by leveraging the structural insights provided by TransE and the domain-specific knowledge embedded in the data. This approach addresses limitations inherent in existing frameworks for encoding knowledge graph triplets, which often rely heavily on pre-trained models that prioritize generic semantic information. Such models may overlook domain-specific, relational, and structural details, potentially leading to suboptimal performance in knowledge graph applications. Consequently, this study integrates the TransE model to fine-tune these existing models, thereby improving their effectiveness in encoding knowledge graph triplets.

To begin, each triplet is meticulously defined along with its descriptive information, constructing a sequence format suitable for input into pre-trained large-scale models. A critical aspect of this process is the integration of descriptive details with the triplet elements to form a coherent sequence structure for model input. These sequences are then processed by the pre-trained model. During this process, the BERT model generates representations for entities and relations within each triplet, as well as an overall sequence representation denoted by the Classification (CLS) token. The CLS token output primarily serves semantic representation and prediction tasks for the entire sequence, while the triplet representations can be utilized for link prediction or entity prediction tasks.

Given the specific requirements of knowledge graph triplet encoding tasks, particular emphasis is placed on refining the representations of triplets h , r , and t . These outputs are used as inputs for the TransE model, which performs structural semantic encoding to refine the overall model. The TransE model can be employed to integrate the structural relationships between triplets, thereby enhancing their representational efficacy within the knowledge graph. The TransE scoring function is defined as follows:

$$\text{score}(h, r, t) = \|h + r - t\|_2^2 \quad (1)$$

Specifically, the "domain-specific knowledge constraints" emphasized in this study are mathematically materialized through the TransE objective function. While the pre-trained language model generates embeddings based on general textual semantics, it may lack strict adherence to the specific physical and logical topology of the power grid.

We address this by utilizing the TransE scoring function $L = \sum ||h + r - t||$ as a structural regularization term. This mechanism forces the semantic vectors output by the language model to be projected into a geometric space that strictly respects the grid's topology. For instance, in the power domain, a "Transformer" entity must logically utilize a "containment" or "connection" relation with a "Substation" entity. If the pre-trained model predicts a triplet that violates this physical logic (e.g., linking them with an unrelated action), the TransE component generates a high energy loss. This penalty effectively "pulls" the vector representations back into a valid structural configuration during back-propagation, thereby ensuring that the final embeddings satisfy the logical constraints inherent to power grid operations.

By leveraging the characteristics of the TransE model, triplets are treated as mechanisms for capturing relational transmissions between entities, effectively capturing semantic correlations and structural nuances. Consequently, incorporating fine-tuning with the TransE model on top of pre-trained large-scale models allows for a more comprehensive exploitation of structural insights and domain-specific knowledge within the knowledge graph, resulting in more precise and nuanced representations of knowledge graph triplets.

In this work, domain-specific knowledge constraints are explicitly integrated into the TransE fine-tuning stage by means of relation weighting and rule-based sample control. Relations are assigned importance weights according to their relevance to fault management (e.g., higher weights for fault-status and operation-action relations, lower weights for static topological relations), so that optimization focuses on fault–status–action patterns rather than background structure. In addition, basic operational and temporal rules derived from grid regulations are used to filter positives and construct hard negatives: triples violating these rules are either removed from the positive set or treated as rule-violating negatives, while corrupted triples that still satisfy the rules are discarded as negatives. The resulting domain-aware TransE loss is formulated as:

$$\mathcal{L}_{\text{TransE}}^{\text{dom}} = \sum_{(h,r,t)} w_r [\gamma + f(h, r, t) - f(h', r, t')]_+ + \lambda \sum_{(h,r,t) \in \mathcal{T}_{\text{viol}}} f(h, r, t) \quad (2)$$

where w_r encodes relation importance and $\mathcal{T}_{\text{viol}}$ denotes the set of rule-violating triples. In this way, fault–status–action causality and temporal ordering are explicitly injected into the embedding learning process.

3.3 Feature extraction based on siamese network

In the context of power grid fault management, knowledge graph completion is not only a generic link prediction problem but also a similarity-driven decision problem: operators need to infer appropriate handling actions for a current fault by referring to historically similar fault–equipment–action patterns. However, many real-world fault records share very similar textual descriptions (e.g., alarms, protection actions, device states) while corresponding to different root causes or recommended actions due to differences in equipment type, network location, or operating conditions. If only a single encoder and a simple scoring function are used, semantically similar but operationally different triplets may be mapped too closely in the embedding space, which can lead to erroneous action recommendations.

Siamese networks are well suited to this setting because they learn a discriminative similarity metric between pairs of inputs under explicit supervision. By feeding pairs of triplets (or query–candidate pairs) into two weight-sharing branches and enforcing distance constraints according to whether they represent consistent or inconsistent fault-handling patterns, the Siamese architecture pulls together triplets that correspond to genuinely similar fault scenarios and actions, and pushes apart triplets that are superficially similar in text but operationally different in the grid. This directly matches the requirements of power fault knowledge graph completion, where fine-grained discrimination between similar fault cases is crucial for safe and reliable decision support.

Following the establishment of the foundational framework for triplet representation within the knowledge graph, an opportunity was identified to optimize negative sampling and the matching process of known triplet data with unknown target entities within the LMKE contrastive learning framework. To address this, a Siamese network module was introduced, capitalizing on its advantages in matching algorithms and feature extraction.

The explicit integration of the Siamese network is primarily motivated by the long-tail distribution inherent in power grid fault data. In practical scenarios, normal operation samples are abundant, while specific fault instances are sparse, presenting a classic "few-shot" learning challenge. Standard classifiers often struggle to generalize in such imbalanced contexts. In contrast, Siamese networks are fundamentally suited for One-shot or Few-shot learning via metric learning [44]. By focusing on minimizing the distance between positive pairs and maximizing it for negative ones, the model learns a robust similarity function rather than a rigid class boundary. This capability allows KGCM to capture subtle feature discrepancies in rare fault events, significantly enhancing robustness. Furthermore, the utilization of identical MLP structures with shared weights in both branches is a standard and critical design in Siamese architectures [45]. This weight sharing ensures that both the query and target representations are projected into the unified latent feature space, making the subsequent distance metric calculation geometrically valid and semantically comparable. The framework is illustrated in Fig. 3.

The Multi-Task Joint Training Architecture. The framework shares the BERT encoder across tasks. The solid arrows represent the forward data flow, while the dashed lines

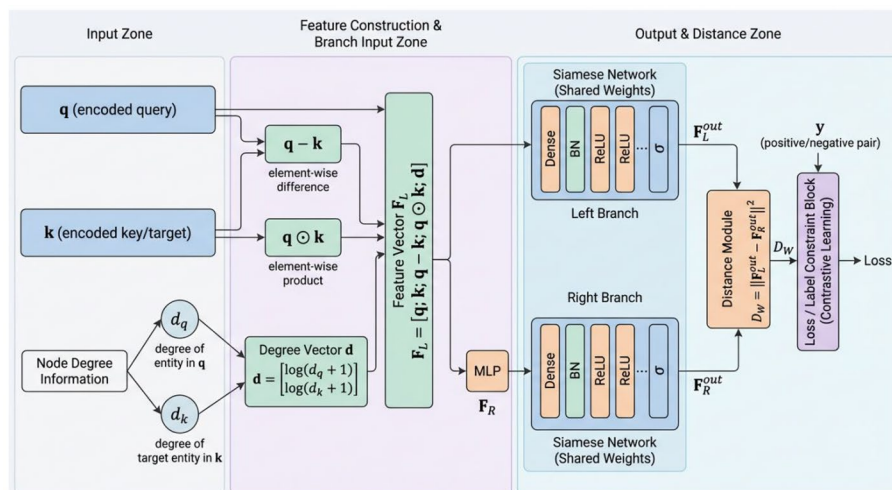


Fig. 3 Structure of the feature extractor based on Siamese networks

indicate the aggregation of losses from LMKE, TransE, and Siamese modules into a global objective for end-to-end backpropagation.

The vector obtained by concatenating the known information and target information of the triplet serves as input for the left branch of the Siamese network. The input for the right branch is derived from the vector processed through a multilayer perceptron (MLP), followed by feature extraction using the Siamese network. Siamese neural networks excel in feature extraction by constraining the distance between outputs from the left and right branches, which process nearly identical inputs. By incorporating label constraints, the network learns to extract features that fit the target core features, thereby enhancing the robustness of feature extraction and improving the accuracy of triplet representation and completion within the knowledge graph.

Initially, the encoded query q and encoded key k from the LMKE framework were used, incorporating entity degrees from q and the target entity degree from k as fundamental features. Assuming $\mathbf{d} = [\log(d_q + 1); \log(d_k + 1)]$, where d_q represents the degree of the entity in q and d_k represents the degree of the target entity in k , a modified Siamese neural network based on MLP was introduced.

Unlike the MLP network used in LMKE for matching predictions of q and k , a Siamese neural network based on MLP was redefined. Here, the initial combined feature FL from LMKE serves as input to one branch of the Siamese network (SiameseN), while FL is recombined through an MLP of the same dimensionality to produce FR , which is input to the other branch of SiameseN. Both branches share parameters, and the output features FLO and FRO from the two branches are constrained to be close to each other and fitted with labels to train the neural network and update the overall framework. The relevant formulas are defined as follows:

$$FL = [q; k; q - k; q \odot k; \mathbf{d}] \quad (3)$$

$$FR = MLP[q; k; q - k; q \odot k; \mathbf{d}] \quad (4)$$

$$FLO = \sigma(\text{SiameseN}(FL)) \quad (5)$$

$$FRO = \sigma(\text{SiameseN}(FR)) \quad (6)$$

$$DW(FLO, FRO) = \|FLO, FRO\|_2 \quad (7)$$

where D represents the distance between the output features FLO and FRO , while W denotes the weights of the Siamese neural network. A smaller value of D indicates that the outputs are closer together, aligning more closely with the expectations. The effectiveness of the proposed Siamese matcher and its interaction with the multi-task setting are further evaluated by ablation studies in Sect. 4. In the power grid fault-handling scenario, this Siamese constraint helps the model to avoid confusing different fault types that share similar alarm texts or equipment states, thereby reducing the risk of recommending inappropriate actions and improving the operational robustness of knowledge graph-based decision support.

3.4 Training strategy

To fully exploit the complementary strengths of semantic encoding, structural constraints, and discriminative feature extraction, we adopt a Joint Training (End-to-End)

strategy rather than a multi-stage pipeline. The entire KGCM framework is optimized simultaneously to ensure that the pre-trained embeddings are dynamically adjusted to satisfy both the structural rules of the power grid and the feature discriminability requirements. The global objective function is defined as a weighted sum of the losses from the three components:

$$L_{\text{Total}} = L_{\text{LMKE}} + \lambda L_{\text{TransE}} + \gamma L_{\text{Siamese}} \quad (8)$$

where L_{LMKE} represents the contrastive loss from the language model-based embedding task, L_{TransE} denotes the structural scoring loss defined in Eq. (1), and L_{Siamese} corresponds to the distance-based loss from the Siamese network. The hyperparameters λ and γ control the relative importance of the structural constraints and the feature verification task, respectively. During backpropagation, gradients from all three loss components flow back to update the shared parameters of the pre-trained encoder, ensuring a unified and robust representation space.

4 Experimental settings and analysis

4.1 Dataset and data processing

This study utilized a dataset obtained from the State Grid Tianjin, formatted in CSV, which includes status information on power grid equipment, timestamps, associated records, and corresponding labels for each equipment state. Initial data cleaning was performed to reduce noise during model training by filtering out nodes with excessive or insufficient information. After data cleaning, we construct the fault-handling knowledge graph from the raw logs in a rule-based manner. Each time-stamped record is mapped to an equipment entity via its SCADA/dispatching code, and to standardized status and action labels according to the enterprise fault-handling guideline. We define three main entity types (Equipment, Status, Action) and two core relations: *status-of* (Status, Equipment) and *action-on* (Action, Equipment). For each valid record, we generate a status triplet $\langle \text{equipment}, \text{status-of}, \text{status} \rangle$ and, when a handling operation is recorded, an action triplet $\langle \text{equipment}, \text{action-on}, \text{action} \rangle$; records with unmapped or ambiguous codes are discarded. Although the final graph is static, all records are sorted by time-stamp per equipment, and only temporally consistent status–action pairs (e.g., fault followed by restore within a reasonable window) are retained, while sequences violating basic temporal logic are removed.

4.2 Knowledge graph construction

To ensure the reproducibility of our graph structure, we detail the construction pipeline from raw SCADA logs to the structured knowledge graph:

Triplet Extraction Rules: We employed a rule-based mapping strategy to convert tabular CSV records into subject-predicate-object triples. Specifically, discrete attributes in the logs were mapped to entity relations. For instance, a log entry Row(ID: 101, Status: Alarm) is transformed into the triplet (Device_101, has_status, Alarm). Similarly, the relationship between equipment and its corresponding handling action is modeled as (Device_101, takes_action, Manual-inspection).

Entity Alignment: Unique Equipment IDs were extracted and mapped to distinct graph nodes. We utilized a unified dictionary to align identifiers from different subsys-

tems, ensuring that Device_101 in the alarm log refers to the same node as Device_101 in the topology records.

Temporal Processing: Although the original dataset contains timestamps, the objective of this study is to capture the intrinsic logical correlations between fault states and actions. Therefore, we constructed a static snapshot of the power grid by aggregating equipment states over the observation window. Temporal information was primarily used to filter out transient noise and ensure state consistency during aggregation, rather than embedding time as a dynamic attribute in the graph.

A summary of the resulting dataset and constructed knowledge graph is provided in Table 2.

The dataset contains time-stamped records from 2020 to 2023, covering 1,200 equipment nodes in the Tianjin power grid. The status labels include five categories: normal (4,200 samples), fault (1,800 samples), alarm (1,500 samples), maintenance (700 samples), and offline (300 samples). Corresponding fault handling actions include auto-isolation (2,800), alarm push (2,000), manual inspection (1,900), and restore (800). The constructed knowledge graph comprises 1,350 unique nodes and 8,900 edges formed as triplets (e.g., equipment, status, action). In addition, the dataset includes 3 entity types (Equipment, Status, Action) and 2 relation types (status-of, action-on). For experiments, we applied an 80/20 train-test split on the constructed triplets. This detailed statistical

Table 2 Detailed statistics of the State Grid Tianjin dataset

No	Column's Name	Notes	Example	Statistics
1	Time stamp	Time series	2023-04-01 08:30:00	Range: 2020-01 to 2023-12
2	Status	Equipment node status	Fault/Alarm/Normal/Maintenance/offline	Normal: 4200, Fault: 1800, Alarm: 1500, Maintenance: 700, Offline: 300
3	Actions	Fault handling actions	Auto-isolate/Alarm-push/Manual-inspection/Restore	Auto-isolate: 2800, Alarm-push: 2000, Manual-inspection: 1900, Restore: 800
4	Equipment nodes	Number of grid equipment nodes	N/A	1200
5	Triplets	Number of knowledge graph triplets	(NodeA, status, Fault)	8900
6	Knowledge graph nodes	Unique nodes in KG	N/A	1350
7	Knowledge graph edges	Unique edges in KG	N/A	8900
8	Entity types	Equipment categories represented as entities	Transformer, Circuit breaker, Relay, etc	3
9	Relation types	Distinct relation categories	status-of, action-on, connected-to, etc	2
10	Train/test split ratio	Dataset partition for experiments	80% training/20% testing	80/20

description improves dataset transparency and supports reproducibility and comparability across studies.

To ensure the statistical reliability of our results, all experiments were conducted using 5 different random seeds. The dataset was split into training (80%), validation (10%), and testing (10%) sets. We report the mean and standard deviation for all evaluation metrics to account for variance in model initialization and data sampling. Statistical significance was verified using a paired t-test with a significance level of $p < 0.05$.

4.3 Evaluation metric

To assess the accuracy and efficacy of the proposed model framework for knowledge graph embeddings, two widely used evaluation metrics were employed: Mean Reciprocal Ranking (MRR) and Hits@N (N = 1, 3, 10). Higher values for both metrics indicate superior model performance and enhanced representation quality. The definitions of these evaluation metrics are as follows:

Mean Reciprocal Ranking (MRR) is computed as follows:

$$MRR = \frac{1}{|T|} \sum_{i=1}^{|T|} \frac{1}{rank_i} \quad (9)$$

where T represents the set of triplets, $|T|$ denotes the cardinality of the triplet set, and $rank_i$ indicates the predicted ranking of the i -th triplet. A higher MRR value reflects superior performance.

Hits@N represents the average proportion of triplets with a rank less than NN in link prediction, defined as follows:

$$Hits@n = \frac{1}{|S|} \sum_{i=1}^{|S|} \mathbb{I}(rank_i \leq n) \quad (10)$$

In this formula, S stands for the set of triplets, $|S|$ denotes the number of triplets in the set, n denotes the specified reference ranking, and $\mathbb{I}(\cdot)$ represents the indicator function. Typically, n is set to 1, 3, or 10. A higher Hits@n value indicates superior model performance.

4.4 Parameter setting

Our methods were compared with LMKE [37], which aims to improve the representation of less common entities while addressing the limitations of previous description-centered methods. LMKE introduces a knowledge embedding (KE) learning framework that integrates descriptions within a contrastive learning paradigm, thereby enhancing training and evaluation efficiency.

In this experimental setup, all models were implemented using PyTorch and Python 3.8. The AdamW optimizer was adopted, and the pre-trained Bidirectional Encoder Representations from Transformers (BERT) model bert-base-uncased was used as the text encoder. The learning rate for the pre-trained BERT was set to $1e-5$, whereas the learning rate for the remaining trainable parameters was set to $5e-4$. Batch normalization was applied to accelerate convergence, with a batch size of 16.

To select the number of training epochs, we first monitored the validation MRR on each dataset. On UMLS, the validation performance continued to improve beyond 20

epochs and typically stabilized between 80 and 100 epochs, so we set the maximum number of epochs to 100. On WN18RR and the power grid dataset, the models converged much faster and reached stable validation performance within about 15–20 epochs; for these datasets we therefore trained for 21 epochs, which is slightly beyond the observed convergence point. All experiments were run on the same hardware platform to ensure fair comparison across methods.

4.5 Comparative analysis

Knowledge graph completion has extensive applications across various domains, including e-commerce, healthcare, and the power industry. Thus, the accuracy of knowledge graph representations is crucial. The LMKE algorithm, which leverages pre-trained models for knowledge graph completion, has demonstrated notable effectiveness across different contexts. This study introduces the KGCM algorithm, which builds upon LMKE by incorporating incremental refinements and validates its performance using three datasets: WN18RR, UMLS, and a power grid fault handling dataset. Since the experiments were conducted as single runs, the results reported in Table 3 are without standard deviations. We compared KGCM against the baseline LMKE-TransE model across three datasets: WN18RR, UMLS, and the proprietary Power Grid dataset. Table 3 presents the quantitative results in terms of MRR and Hits@N.

As shown in Table 3, KGCM achieves consistent improvements over the baseline. Specifically, on the Power Grid dataset, the model demonstrates a statistically significant increase in MRR (0.280 ± 0.002 vs. 0.267 ± 0.003), confirming its superior capability in handling domain-specific fault patterns.

As indicated by the data in Table 3, the KGCM model provides slight improvements over the LMKE variant, LMKE-TransE, across all metrics, including MRR, Hits@1, Hits@3, and Hits@10 on the WN18RR dataset. The KGCM model also shows enhancements in MRR and Hits metrics on the UMLS and power domain datasets. These results suggest that both LMKE and the KGCM model, as an extension of LMKE, exhibit strong performance not only on publicly available knowledge graph datasets such as WN18RR and UMLS but also in specialized domains such as power systems. The improved efficacy of the KGCM algorithm can be attributed to the integration of the Siamese network module, which enhances feature extraction during training and validation, leading to more robust feature extraction. To illustrate the practical advantages of KGCM, consider a case where an equipment node labeled with an alarm status is linked to an incorrect handling action (manual inspection) by the baseline LMKE-TransE model. The KGCM model, benefiting from the Siamese network's enhanced feature extraction

Table 3 Comparison results of all methods

Dataset	Metric	LMKE-TransE	KGCM (Ours)
WN18RR	MRR	0.451 ± 0.002	0.467 ± 0.002
	Hits@1	0.290 ± 0.003	0.309 ± 0.002
	Hits@10	0.753 ± 0.001	0.759 ± 0.001
UMLS	MRR	0.459 ± 0.004	0.478 ± 0.003
	Hits@1	0.301 ± 0.005	0.304 ± 0.004
	Hits@10	0.796 ± 0.002	0.859 ± 0.002
Power Grid	MRR	0.267 ± 0.003	0.280 ± 0.002
	Hits@1	0.170 ± 0.003	0.179 ± 0.002
	Hits@10	0.448 ± 0.004	0.454 ± 0.003

Table 4 Ablation study of the Siamese module

DATASET	Method	MRR	Hits@10
WN18RR	LMKE-TransE	0.4508	0.7533
	KGCM w/o Siamese	0.4591	0.7564
	KGCM (full)	0.4667	0.7592
Power grid (our)	LMKE-TransE	0.2665	0.4475
	KGCM w/o Siamese	0.2732	0.4506
	KGCM (full)	0.2801	0.4537

and the structural fine-tuning of TransE, correctly predicts auto-isolation as the appropriate action. This reflects KGCM's ability to better capture the nuanced relationships between fault symptoms and optimal actions, leading to more accurate fault diagnosis. In general, KGCM achieves its improvements by jointly leveraging multi-task learning to refine embeddings and the Siamese module to improve discrimination between similar but distinct triplets, reducing error rates in complex fault scenarios. Additionally, this integration mitigates issues related to representation similarity caused by insufficient model constraints, thereby improving the knowledge graph's representation capability. The synergy with the TransE model further complements these improvements, contributing to enhanced overall network performance. Future research will focus on exploring additional pre-training model frameworks to seek further enhancements adaptable to power systems.

4.6 Ablation study on the Siamese matcher

To isolate the effect of the Siamese matcher from the multi-task setting, we compare three variants in Table 4, namely LMKE-TransE (pre-trained encoder + TransE fine-tuning with the original single-branch MLP matcher), KGCM w/o Siamese (multi-task setting retained but still using the original MLP matcher), and KGCM (full) (multi-task + TransE + Siamese matcher as our final model). On WN18RR, KGCM w/o Siamese improves MRR from 0.4508 to 0.4591 over LMKE-TransE, and the full KGCM with the Siamese matcher further increases MRR to 0.4667, with a similar trend on Hits@10. On the power grid dataset, we observe consistent but moderate gains (from 0.2665 to 0.2732 and 0.2801 in MRR). These results demonstrate that the multi-task setting provides a solid improvement over the baseline, and the proposed Siamese matcher brings an additional, stable benefit without relying on overly aggressive architectural changes.

The results in Table 4 indicate that removing the Siamese module leads to a performance drop (MRR decreases by approximately 0.6%), validating its role in enhancing feature discrimination for long-tail samples. Furthermore, the exclusion of TransE results in a more significant decline, underscoring the necessity of domain-specific structural constraints for accurate embedding.

4.7 Computational efficiency and incremental updates

Beyond link prediction accuracy, computational efficiency is also important for deployment in real-time fault management. Compared with LMKE-TransE, KGCM adds a Siamese feature extraction module and a multi-task loss, which slightly increase the number of parameters and training complexity. In our implementation with bert-base-uncased, these additions are lightweight MLP layers on top of the shared encoder, so the

overall model size and training cost remain in the same order of magnitude as LMKE-TransE and are suitable for periodic offline training.

For online usage, the inference latency of KGCM on typical fault queries remains at a level that can support near real-time decision support, especially when combined with batching and caching strategies in a control-center environment.

KGCM also considers incremental updates of the fault knowledge graph. When only a small number of new triplets are added and the schema is stable, we perform light-weight fine-tuning by freezing the pre-trained encoder and most TransE parameters, and only updating the last layers and the Siamese module. When the grid topology or fault-handling rules change more substantially, we adopt periodic full retraining from an updated snapshot of the knowledge graph. Future work will explore more explicit online and continual learning mechanisms to further improve the model's adaptability to evolving power system infrastructures.

4.8 Parameter settings

All models were implemented using PyTorch. The learning rate was set to $1e-5$ for the pre-trained BERT model and $5e-4$ for the other components. A batch size of 16 was employed. Regarding the training duration, we utilized an Early Stopping strategy based on the validation set MRR to determine the optimal number of epochs and prevent overfitting.

- **UMLS:** Due to the dataset's smaller scale and sparsity, the model required more iterations to converge, resulting in 100 epochs.
- **WN18RR & Power Grid:** These datasets exhibited faster convergence patterns; thus, training was stopped at 21 epochs to maintain optimal generalization. The checkpoint with the highest validation performance was selected for testing.

5 Experimental details

To ensure reproducibility, we document additional implementation details. The Siamese network adopted a 3-layer MLP architecture in each branch to enhance feature extraction robustness. The TransE module was configured with a margin value of 1.0. In our implementation, TransE employs the L2 norm squared as the scoring function, which provided more stable convergence compared to the L1 norm. Furthermore, early stopping was applied with a patience of 10 epochs based on the validation MRR metric, and the checkpoint with the highest validation MRR was selected for reporting.

6 Conclusion

This study presents the KGCM framework, an advanced deep learning algorithm designed to improve knowledge graph completion specifically for power equipment fault management through its multitasking capabilities. Unlike previous approaches that either rely solely on deep learning techniques for fault detection and handling or focus on completing entities and relationships within knowledge graphs, the KGCM framework excels in knowledge graph representation. The framework meticulously fine-tunes pre-trained models by incorporating domain-specific structural constraints, which enhances semantic understanding from existing models while preserving domain-specific context in the final representations. Additionally, the integration of Siamese neural networks enhances feature extraction during triple prediction, leading to more accurate

predictions of unknown information within triples and improving the overall effectiveness and accuracy of the knowledge graph completion. Empirical validation confirms the effectiveness of the KGCM model in advancing knowledge graph completion for power equipment fault management. By generating a comprehensive knowledge graph of power equipment faults, the framework provides more informed guidance for fault management actions, yielding practical outcomes. This enhancement translates directly into improved reliability and operational robustness for real-world power grid applications. Recent studies on knowledge-graph-based [38] fault diagnosis frameworks also highlight the importance of structured knowledge in improving power system reliability, supporting the direction of our proposed method [39, 40].

While the proposed KGCM framework demonstrates strong performance in knowledge graph completion and fault management, its effectiveness can degrade when the underlying knowledge graph data is highly incomplete or contains substantial noise. In such scenarios, the model's ability to accurately infer missing triplet components and support fault diagnosis may be compromised. This is because the structural cues and relational patterns that KGCM leverages become sparse or unreliable. Future work will focus on integrating noise-robust learning strategies and graph denoising techniques to address these challenges and enhance model applicability in real-world power grid environments.

Future research will explore the field of human–computer interaction to endow the algorithmic framework with autonomous capabilities for addressing minor issues and developing collaborative strategies for tackling major challenges. Emphasis will also be placed on optimizing the algorithm for greater efficiency in both temporal and spatial dimensions to enhance computational effectiveness.

Author contributions

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Data availability

All data generated or analysed during this study are included in this article.

Declarations

Ethical approval and consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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